

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M. SHAK THJ 192312538 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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18

Student 17: M SHAKTHI | Register Number: 192312538

1. A university must assign 3 professors (P1, P2, P3) to 3 courses (AI, Data Structures, Networks) offered in 3 different time slots such that:

Constraints:

- i. P1 cannot teach in the Networks slot
- ii. P2 must be scheduled before P3 (AI < Data Structures < Networks)
- iii. Only one professor per course/slot
- iv. P3 cannot teach AI
- v. Every professor must be assigned exactly one course

Apply Backtracking Search to find a valid timetable, showing intermediate steps and constraint checks. State the final valid schedule.

2. An autonomous forklift operates in a 5x5 warehouse grid from Start (S) to a loading dock (G), navigating around shelving-unit obstacles and narrow-aisle cells with higher traversal cost.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Normal-aisle move cost = 1, narrow-aisle move cost = 2
- iii. Cannot pass through shelving (obstacle) cells
- iv. The forklift must minimize total cost

Using the above constraints and Uniform Cost Search (UCS), show step-by-step node expansion and priority-queue updates, and determine the optimal path and cost.

Backtracking Search - University Timetable

Given:

3 professors : P_1, P_2, P_3

3 courses : AI, DS, Networks

3 time slots : Slot 1 : AI

Slot 2 : DS

Slot 3 : Networks

Constraints:

P_1 cannot teach networks

P_2 must be scheduled before P_3

only 1 professor per course / slot

P_3 cannot teach AI

Every professor must be assigned exactly 1 course

Backtracking steps:

we assign courses to professors one by one.

Step 1: Assign P_1 :

$P_1 \rightarrow \text{networks} \rightarrow \times$

$P_1 \rightarrow \text{AI} \rightarrow \checkmark$

Step 2: Assign P_2

$P_2 \rightarrow \text{AI} \rightarrow \times$

$P_2 \rightarrow \text{DS} \rightarrow \checkmark$

Step 3: Assign P_3

$P_3 \rightarrow \text{AI} \rightarrow \times$

$P_3 \rightarrow \text{DS} \rightarrow \times$

$P_3 \rightarrow \text{networks} \rightarrow \checkmark$

Time slot	Course	Professor
Slot 1	AI	P1
Slot 2	DS	P2
Slot 3	Networks	P3

constraint check
P1 is not teaching networks
P2 is before P3
P3 is not teaching AI

Hence, the above time table is a valid solution obtained using Backtracking search.

2) Uniform Cost Search (UCS) - Autonomous forklift:

Assumptions:

S - start

G - goal

- shelving obstacle

N - narrow aisle

• = normal aisle

normal move cost = 1

narrow aisle move cost = 2

movement allowed : up, down, left, right.

5x5 Grid:

	C1	C2	C3	C4	C5
R1	S	•	#	•	•
R2	•	•	#	N	•
R3	#	•	N	N	•
R4	•	•	#	N	•
R5	•	•	•	•	G

UCS always expands the node having the lowest accumulated path cost.

Start : $s = (1, 1)$, cost = 0

Possible moves: right $\rightarrow (1, 2)$, cost = 1
down $\rightarrow (2, 1)$, cost = 1

Priority Queue : $(1, 2 : 1), (2, 1 : 1)$

expanded : s

Step 2 $\rightarrow$ Expand $(1, 2)$

Cost = 1

new node : $(2, 2) \rightarrow$ cost = 2

priority Queue : $(2, 1 : 1), (2, 2 : 2)$

expanded : $s, (1, 2)$

Step 3 : Expand $(2, 1)$

cost = 1

$(2, 2)$ is already in the queue with cost 2

priority Queue : $(2, 2 : 2)$

expanded : $s, (1, 2), (2, 1)$

Step 4 : Expand $(2, 2)$

cost = 2

move to : $(3, 2) \rightarrow$ cost = 3

priority queue : $(3, 2 : 3)$

Step 5 : Expand $(3, 2)$

cost = 3

$(4, 2) \rightarrow$ cost = 4

$(3, 3)$ is a narrow circle $\rightarrow$ cost = 5

priority queue : $(4, 2 : 4), (3, 3 : 5)$

VCS selects $(4, 2)$ because $4 < 5$.

Step 6: Expand (4,2)

Cost = 4

(5,2) → cost = 5

(4,1) → cost = 5

priority queue: (3,3:5), (4,1:5), (5,2:5)

Step 7: Expand (3,3)

Cost = 5

This is a narrow aisle cell

moving into another narrow cell gives:

(3,4) → cost = 7

priority queue: (4,1:5), (5,2:5), (3,4:7)

Step 8: Continue through the lower normal aisle

The important expansions are: optimal path:

(4,1): 5

(5,2): 5

(5,1): 6

(5,3): 6

(3,4): 7

(5,4): 7

(3,5): 8

(5,5): 8

3
↓
(2,1)
↓
(2,2)
↓
(3,2)
↓
(4,2)
↓
(5,2)
→
(5,3)
→
(5,4)
→
G

Cost Calculation:

Every cell in this optimal path uses a normal aisle

There are 8 movements, each costs with 1.

Therefore, Total cost = 8×1

= 8

RUBRICS FOR CALCULATION:

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

20+20+22+18+13

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